Problem Set 3

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Problem 3.1 (Planning a stratified sample). We *plan* a survey to estimate the population mean of environmental expenditures (`expend`). We want to assess whether stratification by industry group (`code`) can improve precision relative to a simple random sample without replacement (SRS).

We use the data set `sepe.dat` from Problem Set 1. Treat the data as an SRS of $n = 800$ establishments from a population of size $N = 3770$. We focus on industry class 31, that is, the domain U_{31} of size $N_{31} = 650$, identified with `code == 1`. Assume that:

- the planned net sample size is $n = 800$ establishments;
- the anticipated response rate is 80 % in both strata; and
- the within-stratum variances of `empl` and `expend` can be estimated from the existing SRS `sepe.dat`.

a) Allocate the net sample *proportionally* to the two strata. Determine the required gross sample sizes and estimate the variance of the stratified estimator of the population mean of expenditures.

b) Now use a Neyman-type (optimum) allocation, but base the allocation on the within-stratum standard deviations of the auxiliary variable `empl` rather than on `expend`.
   - Determine the required gross sample sizes.
   - Under what conditions is it reasonable to derive the allocation from an auxiliary variable?
   - Estimate the variance of the stratified estimator of mean environmental expenditures under this allocation.

c) Compare the estimated variances from parts a) and b) with the estimated variance under a simple random sample of $n = 800$. Does stratification improve precision?

Problem 3.2 (Weighing Basu's elephants). A circus owns 50 elephants.[1] The circus owner wants to estimate their total weight without having to weigh every elephant. He believes that Timbo is approximately as heavy as an average elephant and proposes weighing Timbo and multiplying his weight by 50, formally

$$\widehat{t}_y^{\text{owner}} = 50 \cdot y_{\text{Timbo}}.$$

A statistician insists on probability sampling. They agree to select one elephant at random, with the following sampling design:

$$p(s) = \begin{cases} 0.99, & \text{if Timbo is selected,} \\ 1/4900, & \text{otherwise.} \end{cases}$$

[1] This problem is adapted from Basu's famous elephant example in survey sampling; see Basu, D. (1971). An Essay on the Logical Foundations of Survey Sampling, Part 1, in: *Foundations of Statistical Inference*. Proceedings of the Symposium on the Foundations of Statistical Inference Prepared Under the Auspices of the René Descartes Foundation and Held at the Department of Statistics, University of Waterloo, Ontario, Canada, from March 31 to April 9, 1970 ed. by Godambe, V. P. and Sprott, D. A. Toronto/ Montreal: Holt, Rinehart and Winston of Canada, pp. 203–233. The population data are taken from the book: Brewer, K. (2002), *Combined Survey Sampling Inference: Weighing Basu's Elephants*, London: Arnold, Appendix B.

The statistician proposes using the Horvitz–Thompson estimator, $\widehat{t}_y^{\mathrm{HT}}$, of the population total.

a) Express the Horvitz–Thompson estimator in terms of the selected elephant's weight and its selection probability. Show that the estimator is design-unbiased.

b) Suppose Timbo is selected and weighs 3 900 kg. Calculate the estimates obtained by the circus owner and the statistician. Discuss the two estimates. Does design-unbiasedness guarantee that an estimator will give a reasonable estimate for the observed sample?

c) Ken Brewer provides the actual weights of all 50 elephants in the file:

```
> elephants <-
    read.table("https://stat.ethz.ch/Teaching/Datasets/cas-das/elephants.dat",
               header = TRUE)
```

We can use these data to investigate the sampling distribution of the HT estimator. Construct the sampling distribution of the HT estimator by calculating its possible values and their probabilities. Also, calculate the design expectation and standard deviation of the HT estimator. Discuss the results.

Problem 3.3 (PPS Sampling and the Horvitz–Thompson estimator). We use the data set `sepe.dat` from Problem Set 1 and treat the $N = 800$ establishments as the finite population U. We want to draw a PPS sample without replacement of size $n = 200$, using the number of full-time employees as the measure of size. The data set contains:

`empl`: number of full-time employees, denoted by x_i. These values are known from a register before sampling;

`expend`: environmental expenditures, denoted by y_i. We treat these values as unknown until an establishment is sampled.

a) Compute the first-order inclusion probabilities and draw a PPS sample of $n = 200$ establishments without replacement using Brewer's method. If there were no certainty units, the inclusion probabilities would be

$$\pi_i = \frac{n x_i}{t_x}, \qquad t_x = \sum_{i \in U} x_i.$$

Compare these naive probabilities with the probabilities returned by `pps_probabilities()` of the `robsurvey` package. What do you observe?

> **Hint.** Attach the packages `survey` and `robsurvey` to the search path. Compute the inclusion probabilities and draw the sample as follows:
>
> ```
> pps_setup <- pps_probabilities(size = sepe$empl, n = 200)
> pi <- as.numeric(pps_setup)
>
> set.seed(11)
> s <- pps_draw(pps_setup, method = "brewer")
> ```

b) Create a survey design object for the sample from part a). Then use `svyplot()` to plot environmental expenditures against the number of employees. The bubble sizes represent the sampling weights $w_i = 1/\pi_i$. Describe the main features of the plot.

Remarks. For the PPS design of part a), use

```
dn_sepe <- svydesign(ids = ~1, fpc = ~pi, pps = "brewer",
                     data = sepe[s, ])
```

Here, `fpc = ~pi` supplies the first-order inclusion probabilities. The option `pps = "brewer"` tells `survey` to use Brewer's approximation for the second-order inclusion probabilities needed for variance estimation. This is distinct from Brewer's sampling algorithm used in part a).

c) Estimate the population total of environmental expenditures,

$$t_y = \sum_{i \in U} y_i,$$

using the Horvitz–Thompson estimator and the design from part b). Also compute an estimate of the standard error of the estimator.